

Block cipher modes of operation:-

How to securely encrypt and decrypt large amount of data using a block cipher.

Electronic Code Book (ECB)

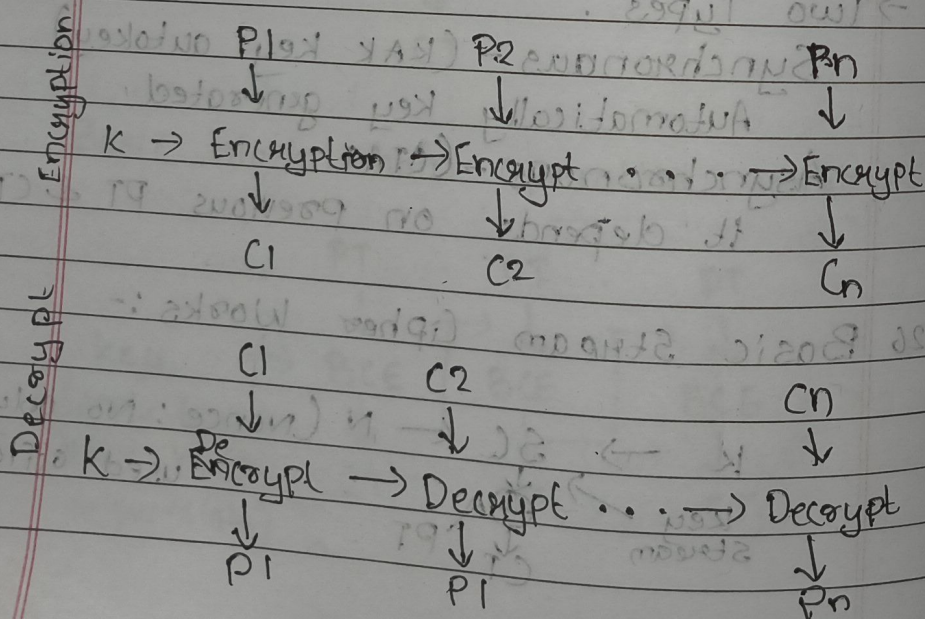
Cipher block chaining (CBC)

Cipher feedback Mode (CFB)

Output Feedback Mode (OFB)

Counter Mode (CM) (CTR)

Electronic Code Book



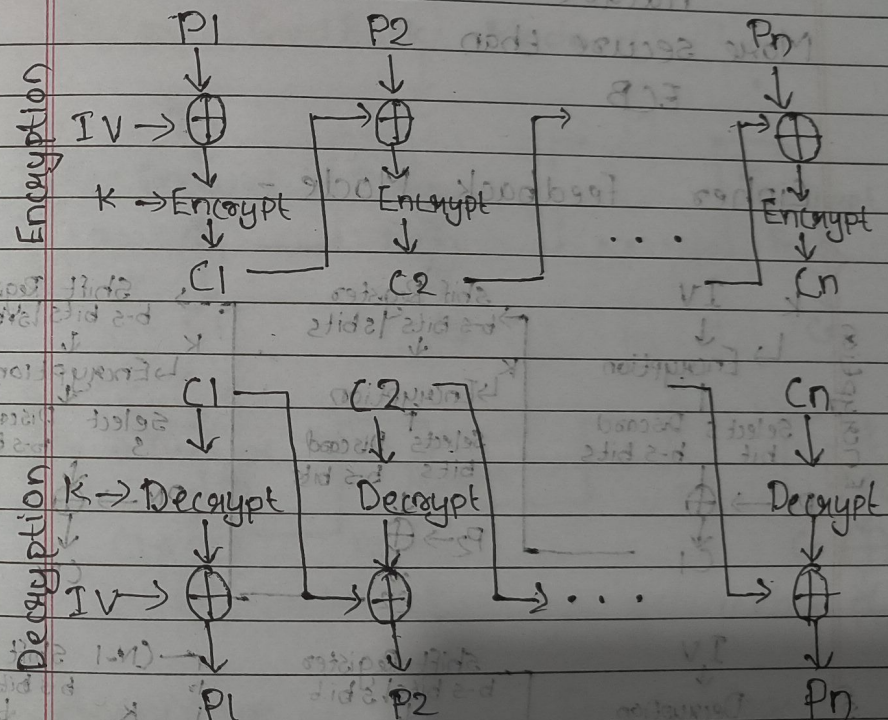
Advantage

Parallel encryption
Faster & Simple way

Disadvantage

Cryptanalysis occurs
Same PT block

Cipher Block Chaining -

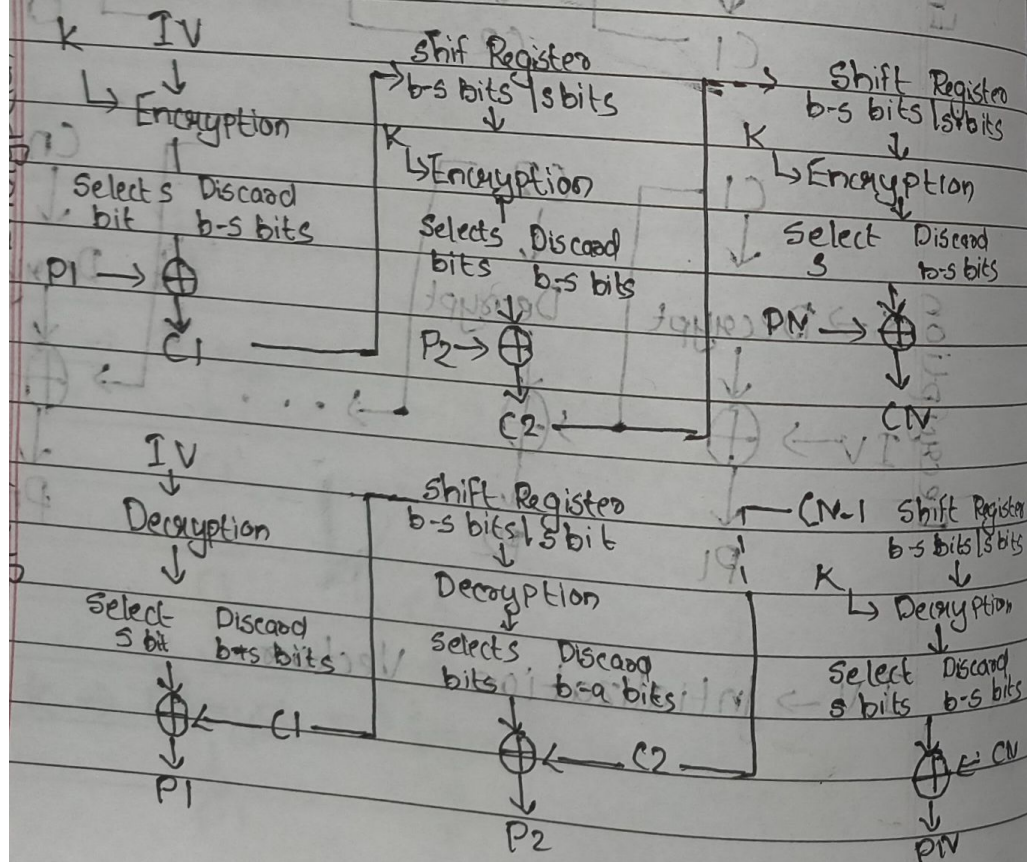


IV → Initialization Vector

Advantages
 Good authentication mechanism
 CBC Work well
 Better resistive to energy nature
 More secure than ECB

Disadvantage
 Previous CI block is different for parallel processing

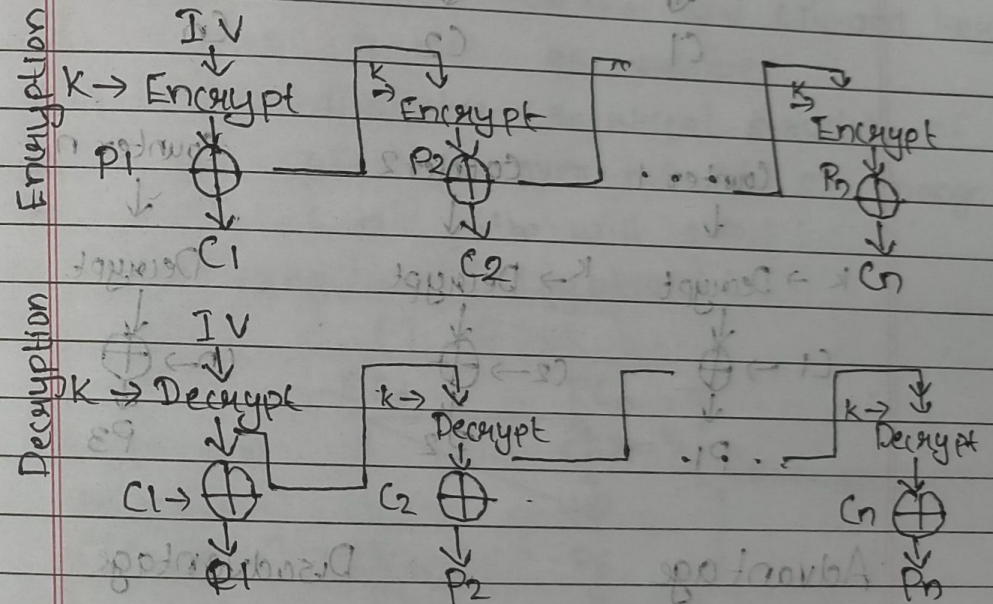
Cipher Feedback Mode -



Advantage
 Crypt analysis is difficult
 It can handle any size of data

Disadvantage
 Parallel processing is not possible
 Complex, more chance of error.

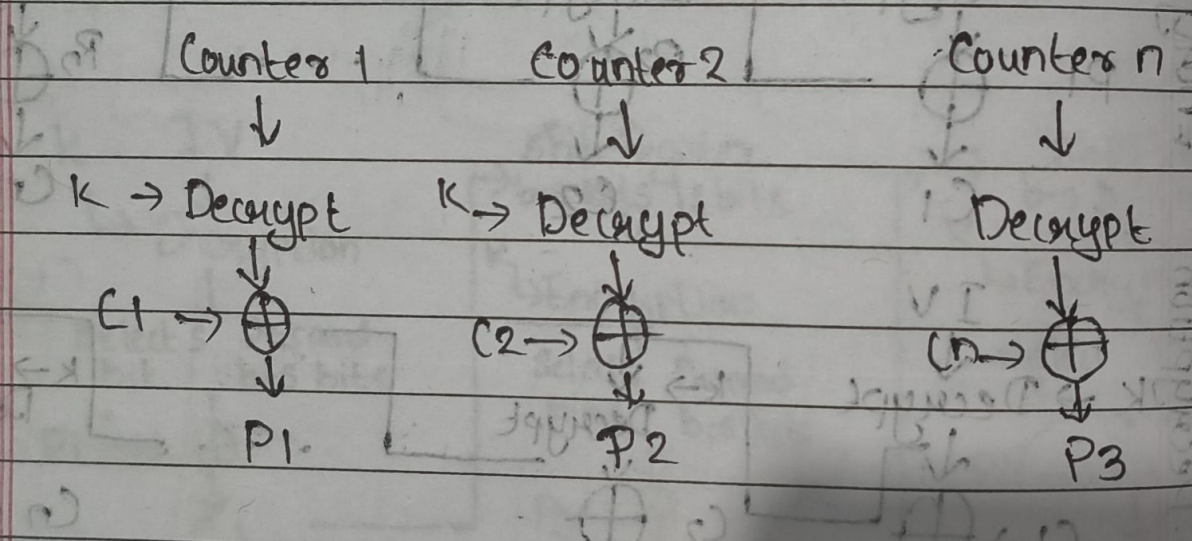
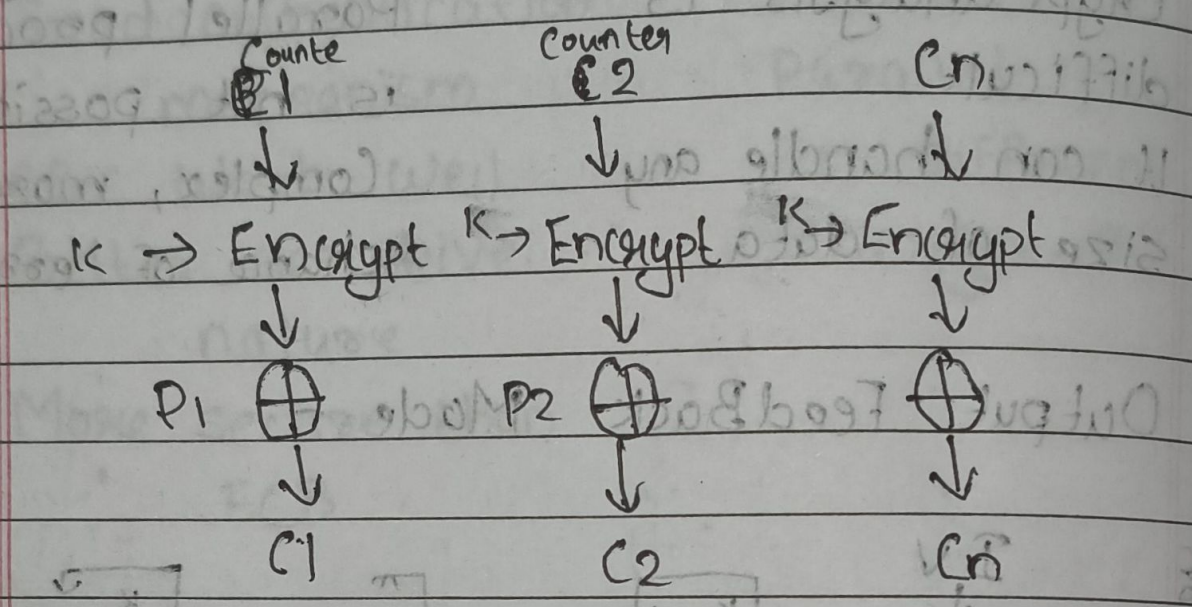
Output Feedback Mode -



Advantage
 Less error propagation

Disadvantage
 OFB is simple to attack than CFB
 Key stream is used

Counter Mode -



Advantage

Disadvantage

Same PT can map to different CT
Parallel execution of encryption

The recovery of PT is inaccuracy when Synchronization is less